

# Visual Studio 2022 – Git Step■by■Step Lab (Text Only)

Beginner■friendly walkthrough for cloning, branching, committing, pushing, pull requests, and resolving merge conflicts using Visual Studio 2022.

## 1) Set Up and Explore Git UI

1. Open Visual Studio 2022.
2. From the menu bar, click View → Git Changes to open the Git Changes window.
3. Click View → Git Repository to open the Git Repository window.
4. Locate the Git menu on the top menu bar (next to Project, Build, etc.).

## 2) Clone an Existing Repository

1. Go to Git → Clone Repository.
2. Paste the repository URL (e.g., from GitHub). Choose a local path.
3. Click Clone. Visual Studio opens the solution automatically if it finds one.

## 3) Create a New Git Repository (Optional)

1. Create a new project (File → New → Project) or open an existing folder/solution.
2. Go to Git → Create Git Repository.
3. Choose to create locally, and optionally connect to a remote host (GitHub/Azure DevOps).
4. Click Create.

## 4) Create and Switch Branches

1. Open View → Git Repository.
2. Right■click the branch you want to branch from (e.g., main) → New Local Branch From...
3. Name it (e.g., feature/login■ui) and create.
4. Double■click the new branch to switch to it.

## 5) Make Changes and Commit

1. Edit or add a file in Solution Explorer.
2. Open View → Git Changes. Verify your changed files appear.
3. Optionally stage specific hunks/files from the Changes list.
4. Write a clear commit message (e.g., 'Add login button and validation').
5. Click Commit All (local) or Commit All and Push (commit + send to remote).

## 6) Push, Pull, and Sync

1. Use Commit All and Push to publish your commit to the remote.
2. Click Pull to fetch and merge latest changes from the remote.
3. Use the Synchronization indicators (status bar) to see pending changes.

## 7) Create a Pull Request (PR)

1. Right■click your branch in Git Repository → Create Pull Request.
2. A browser window opens for GitHub/Azure DevOps.

3. Fill in title, description, reviewers, and target branch (usually main).
4. Create the PR.

## 8) Compare Files and Branches

1. In Solution Explorer, right-click a modified file → Compare with Unmodified to view file diffs.
2. In Git Repository window, right-click a branch → Compare with Current Branch to view branch diffs.

## 9) Resolve Merge Conflicts

1. If a conflict appears after Pull/Merge/Rebase, open the conflicted file.
2. Use the in-editor controls: Accept Current, Accept Incoming, or Keep Both.
3. Manually edit if needed, then save the file.
4. Return to Git Changes. Mark resolved (staging will mark as resolved).
5. Commit the merge resolution and push.

## Quick Reference & Checklist

| Task                | Where in VS2022                     | You should see                    |
|---------------------|-------------------------------------|-----------------------------------|
| Open Git Changes    | View → Git Changes                  | Changed files, commit message box |
| Open Git Repository | View → Git Repository               | Local/Remote branches, actions    |
| New Branch          | Git Repository → Right-click branch | New Local Branch dialog           |
| Commit              | Git Changes                         | Commit button becomes enabled     |
| Push                | Git Changes                         | Status bar shows outgoing changes |
| Pull                | Git Changes                         | Status bar shows incoming changes |
| Create PR           | Git Repository → Right-click branch | Browser opens PR form             |